

ABOUT THIS BOOK

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

License: <https://creativecommons.org/licenses/by-sa/3.0/>
and

<https://www.gnu.org/licenses/fdl-1.3.en.html>

First manual appeared in January 2018

First edition published on July 10, 2024

In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

CONTENTS

1	GATE CE-1 2026	3
1.1	Formulae	3
1.2	Problems	8

1. GATE CE-1 2026

A. Formulae

1.1.1 Intersection of Conics and Common Chord

- Conic in Vector Form: Every second-degree curve $ax^2 + 2bxy + cy^2 + 2dx + 2ey + f = 0$ can be written in matrix-vector form as:

$$\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0, \quad \mathbf{V} = \begin{pmatrix} a & b \\ b & c \end{pmatrix} = \mathbf{V}^\top, \quad \mathbf{u} = \begin{pmatrix} d \\ e \end{pmatrix} \quad (1.1.1.1)$$

The conic is a parabola if $|\mathbf{V}| = 0$, an ellipse if $|\mathbf{V}| > 0$, and a hyperbola if $|\mathbf{V}| < 0$. For a circle with centre $\mathbf{c}$ and radius r , $\|\mathbf{x} - \mathbf{c}\|^2 = r^2$ gives $\mathbf{V} = \mathbf{I}$, $\mathbf{u} = -\mathbf{c}$, and $f = \|\mathbf{c}\|^2 - r^2$.

- Line-Conic Intersection: The points of intersection of a line $L : \mathbf{x} = \mathbf{h} + \kappa \mathbf{m}$ ($\kappa \in \mathbb{R}$) with a conic section $\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0$ are given by:

$$\mathbf{x}_i = \mathbf{h} + \kappa_i \mathbf{m} \quad (1.1.1.2)$$

where the parameter values κ_i are:

$$\kappa_i = \frac{-\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u}) \pm \sqrt{[\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u})]^2 - g(\mathbf{h}) (\mathbf{m}^\top \mathbf{V} \mathbf{m})}}{\mathbf{m}^\top \mathbf{V} \mathbf{m}} \quad (1.1.1.3)$$

with $g(\mathbf{h}) = \mathbf{h}^\top \mathbf{V} \mathbf{h} + 2\mathbf{u}^\top \mathbf{h} + f$. The line meets the conic at two, one, or no real points according as the discriminant

$$\Delta = [\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u})]^2 - g(\mathbf{h}) (\mathbf{m}^\top \mathbf{V} \mathbf{m}) \quad (1.1.1.4)$$

is positive, zero, or negative.

- Derivation of the Intersection Parameter: Substituting $\mathbf{x} = \mathbf{h} + \kappa \mathbf{m}$ into the conic equation and using symmetry $\mathbf{V} = \mathbf{V}^\top$:

$$(\mathbf{h} + \kappa \mathbf{m})^\top \mathbf{V} (\mathbf{h} + \kappa \mathbf{m}) + 2\mathbf{u}^\top (\mathbf{h} + \kappa \mathbf{m}) + f = 0 \quad (1.1.1.5)$$

$$\kappa^2 (\mathbf{m}^\top \mathbf{V} \mathbf{m}) + 2\kappa \mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u}) + g(\mathbf{h}) = 0 \quad (1.1.1.6)$$

When $\Delta = 0$, the two roots coincide and the line is tangent to the conic.

- Pencil of Conics: The intersection of two conics with parameters $(\mathbf{V}_1, \mathbf{u}_1, f_1)$ and $(\mathbf{V}_2, \mathbf{u}_2, f_2)$ lies on every member of the pencil:

$$\mathbf{x}^\top (\mathbf{V}_1 + \mu \mathbf{V}_2) \mathbf{x} + 2(\mathbf{u}_1 + \mu \mathbf{u}_2)^\top \mathbf{x} + (f_1 + \mu f_2) = 0 \quad (1.1.1.7)$$

- Degenerate Conic and Common Chord: The pencil represents a pair of straight lines (a degenerate conic) if and only if:

$$\begin{vmatrix} \mathbf{V}_1 + \mu \mathbf{V}_2 & \mathbf{u}_1 + \mu \mathbf{u}_2 \\ (\mathbf{u}_1 + \mu \mathbf{u}_2)^\top & f_1 + \mu f_2 \end{vmatrix} = 0 \quad (1.1.1.8)$$

If $\mathbf{V}_1 = \mathbf{V}_2$ (e.g., two circles), choosing $\mu = -1$ removes the second-degree terms, leaving the common chord (radical axis):

$$2(\mathbf{u}_1 - \mathbf{u}_2)^\top \mathbf{x} + (f_1 - f_2) = 0 \quad (1.1.1.9)$$

1.1.2 Line Intersection and Angle Between Vectors

- Forms of a Line: A line through point $\mathbf{A}$ with direction vector $\mathbf{m}$ can be written in parametric form or normal form using normal vector $\mathbf{n}$ ($\mathbf{n}^\top \mathbf{m} = 0$):

$$\mathbf{x} = \mathbf{A} + \kappa \mathbf{m}, \quad \mathbf{n}^\top (\mathbf{x} - \mathbf{A}) = 0 \quad (1.1.2.1)$$

In 2D, if $\mathbf{m} = \begin{pmatrix} m_1 \\ m_2 \end{pmatrix}$, then $\mathbf{n} = \begin{pmatrix} -m_2 \\ m_1 \end{pmatrix}$.

- Line Intersection and Solvability: The lines $\mathbf{x} = \mathbf{A} + \kappa_1 \mathbf{m}_1$ and $\mathbf{x} = \mathbf{B} + \kappa_2 \mathbf{m}_2$ meet where:

$$\begin{pmatrix} \mathbf{m}_1 & -\mathbf{m}_2 \end{pmatrix} \begin{pmatrix} \kappa_1 \\ \kappa_2 \end{pmatrix} = \mathbf{B} - \mathbf{A} \quad (1.1.2.2)$$

This system has a unique solution if and only if $|\mathbf{m}_1 \quad \mathbf{m}_2| \neq 0$. If the determinant is zero, the lines are parallel or coincident.

- Angle Between Vectors: The angle θ between two vectors $\mathbf{a}$ and $\mathbf{b}$ is given by:

$$\cos \theta = \frac{\mathbf{a}^\top \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} \quad (1.1.2.3)$$

Two vectors are orthogonal if and only if $\mathbf{a}^\top \mathbf{b} = 0$.

- Circle Representation and Chords: A point on a circle of radius r centered at the origin is $\mathbf{x} = r \begin{pmatrix} \cos \alpha \\ \sin \alpha \end{pmatrix}$ with $\mathbf{e}_\alpha \begin{pmatrix} \cos \alpha \\ \sin \alpha \end{pmatrix}$. The chord length between $\mathbf{x}_\alpha = r \mathbf{e}_\alpha$ and $\mathbf{x}_\beta = r \mathbf{e}_\beta$ satisfies:

$$\mathbf{e}_\alpha^\top \mathbf{e}_\beta = \cos(\alpha - \beta) \quad (1.1.2.4)$$

$$\|\mathbf{x}_\alpha - \mathbf{x}_\beta\|^2 = 2r^2 (1 - \mathbf{e}_\alpha^\top \mathbf{e}_\beta) = 4r^2 \sin^2 \frac{\alpha - \beta}{2} \quad (1.1.2.5)$$

1.1.3 System of Linear Equations and Rouché–Capelli Theorem

- Consistency and Solvability: A system

$$\mathbf{A}\mathbf{x} = \mathbf{b}, \quad \mathbf{A} \in \mathbb{R}^{m \times n} \quad (1.1.3.1)$$

is consistent if and only if

$$\text{rank}(\mathbf{A}) = \text{rank}([\mathbf{A} \mid \mathbf{b}]) = r \quad (1.1.3.2)$$

If $r = n$, the solution is unique; if $r < n$, there exist infinitely many solutions with $n - r$ free variables. If $\mathbf{A} \in \mathbb{R}^{n \times (n-1)}$, the system is consistent only if

$$|\mathbf{A} \quad \mathbf{b}| = 0 \quad (1.1.3.3)$$

- Homogeneous System and Zero Norm Property: For $\mathbf{A} \in \mathbb{R}^{2 \times 3}$ with rank 2, the solution of $\mathbf{A}\mathbf{x} = \mathbf{0}$ is given by the cross product of its rows:

$$\mathbf{x} = \kappa (\mathbf{a}_1 \times \mathbf{a}_2) \quad (1.1.3.4)$$

For any vector $\mathbf{v} \in \mathbb{R}^n$,

$$\|\mathbf{v}\|^2 = \mathbf{v}^\top \mathbf{v} = \sum_i v_i^2 = 0 \iff \mathbf{v} = \mathbf{0} \quad (1.1.3.5)$$

1.1.4 Relation Between Eigenvalues and Trace of a Matrix

- **Characteristic Polynomial and Trace Relation:** The characteristic polynomial of $\mathbf{A} \in \mathbb{R}^{n \times n}$ is

$$P(\lambda) = |\lambda \mathbf{I} - \mathbf{A}| \quad (1.1.4.1)$$

The sum of all eigenvalues equals the trace of the matrix:

$$\sum_{i=1}^n \lambda_i = \sum_{i=1}^n a_{ii} = \text{tr}(\mathbf{A}) \quad (1.1.4.2)$$

A matrix is symmetric if $\mathbf{A}^\top = \mathbf{A}$ and skew-symmetric if $\mathbf{A}^\top = -\mathbf{A}$.

1.1.5 Cholesky Decomposition

- **Factorization:** A symmetric positive definite matrix $\mathbf{A}$ can be factorized as $\mathbf{A} = \mathbf{L}\mathbf{L}^\top$ where $\mathbf{L}$ is lower triangular.
- **2×2 Decomposition Form:**

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{pmatrix} = \begin{pmatrix} l_{11} & 0 \\ l_{21} & l_{22} \end{pmatrix} \begin{pmatrix} l_{11} & l_{21} \\ 0 & l_{22} \end{pmatrix} = \begin{pmatrix} l_{11}^2 & l_{11}l_{21} \\ l_{11}l_{21} & l_{21}^2 + l_{22}^2 \end{pmatrix} \quad (1.1.5.1)$$

1.1.6 Bayes' Theorem in Vector Form

- **Vector Formulation:** Given prior vector $\boldsymbol{\pi} = (\Pr(B_1) \ \dots \ \Pr(B_n))^\top$ and likelihood vector $\mathbf{l} = (\Pr(E | B_1) \ \dots \ \Pr(E | B_n))^\top$:

$$\Pr(E) = \boldsymbol{\pi}^\top \mathbf{l} \quad (1.1.6.1)$$

$$\Pr(B_i | E) = \frac{\pi_i l_i}{\boldsymbol{\pi}^\top \mathbf{l}} \quad (1.1.6.2)$$

1.1.7 Minimization via Completing the Square

- **Inequality and Equality Condition:** For real a and b :

$$ab - b^2 = \frac{a^2}{4} - \left(b - \frac{a}{2}\right)^2 \leq \frac{a^2}{4} \quad (1.1.7.1)$$

with equality holding if and only if $b = \frac{a}{2}$.

- **Zero Integral Property:** If $g(x) \geq 0$ is continuous on $[a, b]$ and $\int_a^b g(x) dx = 0$, then $g(x) = 0$ on $[a, b]$.

1.1.8 Laplace Transform and State-Space Form

- Transform Operational Properties:

$$y'(t) \xleftrightarrow{\mathcal{L}} sY(s) - y(0) \quad (1.1.8.1)$$

$$y''(t) \xleftrightarrow{\mathcal{L}} s^2Y(s) - sy(0) - y'(0) \quad (1.1.8.2)$$

$$e^{at}u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s-a} \quad (1.1.8.3)$$

- State-Space Representation: Writing $\mathbf{z} = \begin{pmatrix} y \\ y' \end{pmatrix}^\top$, the ODE $y'' + py' + qy = 0$ becomes $\mathbf{z}' = \mathbf{A}\mathbf{z}$ where:

$$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ -q & -p \end{pmatrix} \quad (1.1.8.4)$$

The exponents of the solution are the eigenvalues of $\mathbf{A}$.

1.1.9 Newton-Raphson Method

- First-Order Recurrence: For a differentiable function $f(x)$, the root is iteratively approximated by:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad (1.1.9.1)$$

1.1.10 Polynomial Interpolation (Vandermonde Matrix)

- Matrix Form: The polynomial $P_2(x) = c_0 + c_1x + c_2x^2$ through points (x_i, y_i) for $i = 1, 2, 3$ satisfies:

$$\begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} \quad (1.1.10.1)$$

1.1.11 Equilibrium of Rigid Bodies in Vector Form

- Force and Moment Equilibrium: For a body in equilibrium under forces $\mathbf{F}_i$ acting at points $\mathbf{r}_i$:

$$\sum_i \mathbf{F}_i = \mathbf{0}, \quad \sum_i (\mathbf{r}_i - \mathbf{O}) \times \mathbf{F}_i = \mathbf{0} \quad (1.1.11.1)$$

where in the 2D plane, $\mathbf{r} \times \mathbf{F} = r_x F_y - r_y F_x$.

- Bending Moment and Friction: The bending moment at section P is

$$M(P) = \sum_{\text{one side}} (\mathbf{r}_i - \mathbf{P}) \times \mathbf{F}_i \quad (1.1.11.2)$$

Limiting friction force is $F = \mu N$.

1.1.12 Direct Stiffness Method

- Member and Global Stiffness Formulation: For an axial member of length L , area A , modulus E , and direction vector $\mathbf{m}$, the nodal stiffness is

$$\mathbf{k} = \frac{AE}{L} \mathbf{m} \mathbf{m}^\top \quad (1.1.12.1)$$

and the member stiffness matrix is $\begin{pmatrix} \mathbf{k} & -\mathbf{k} \\ -\mathbf{k} & \mathbf{k} \end{pmatrix}$. The global system is

$$\mathbf{K}\mathbf{u} = \mathbf{F} \quad (1.1.12.2)$$

- Thermal Load Equivalence: A temperature rise ΔT with expansion coefficient α produces equivalent nodal loads of $AE\alpha\Delta T$ at member ends directed outward.

1.1.13 Cellular Update Rule in Matrix Form

- Matrix Formulation: For N bulbs in a line with state vector $\mathbf{x}_k \in \{0, 1\}^N$, the number of ON neighbours is given by $\mathbf{s}_k = \mathbf{A}\mathbf{x}_k$ with:

$$a_{ij} = \begin{cases} 1, & |i - j| = 1 \\ 0, & \text{otherwise} \end{cases} \quad (1.1.13.1)$$

1.1.14 Surveying and Traffic Relations

- Reduced Level: With height of instrument HI and staff reading h ,

$$RL = HI - h \quad (1.1.14.1)$$

- Space Mean Speed: For n vehicles covering distance d in times $\mathbf{t} = (t_1 \ \dots \ t_n)^\top$:

$$v_s = \frac{nd}{\mathbf{1}^\top \mathbf{t}} = \frac{n}{\sum_i \frac{1}{v_i}} \quad (1.1.14.2)$$

which is the harmonic mean of the spot speeds $v_i = d/t_i$.

The point T lies beyond R on PR and beyond S on QS . Hence $\overrightarrow{TR}$ is along $\mathbf{P} - \mathbf{R}$ and $\overrightarrow{TS}$ is along $\mathbf{Q} - \mathbf{S}$. Using (1.1.2.3),

$$\cos \angle RTS = \left(\cos \frac{\alpha}{2} \quad \sin \frac{\alpha}{2} \right) \begin{pmatrix} -\sin \frac{\beta}{2} \\ \cos \frac{\beta}{2} \end{pmatrix} \quad (1.1.2.5)$$

$$= \sin \frac{\alpha}{2} \cos \frac{\beta}{2} - \cos \frac{\alpha}{2} \sin \frac{\beta}{2} = \sin \frac{\alpha - \beta}{2} \quad (1.1.2.6)$$

$$= \sin 40^\circ = \cos 50^\circ \quad (1.1.2.7)$$

$$\implies \angle RTS = 50^\circ \quad (1.1.2.8)$$

For a numerical check, take $r = 1$, $\alpha = 130^\circ$, $\beta = 50^\circ$ so that $\alpha - \beta = 80^\circ$. From (1.1.2.2),

$$(\mathbf{R} - \mathbf{P} \quad -(\mathbf{S} - \mathbf{Q})) \begin{pmatrix} \kappa_1 \\ \kappa_2 \end{pmatrix} = \mathbf{Q} - \mathbf{P} \implies \mathbf{T} = \begin{pmatrix} 0 \\ 2.1445 \end{pmatrix} \quad (1.1.2.9)$$

and (1.1.2.3) gives $\angle RTS = 50^\circ$.

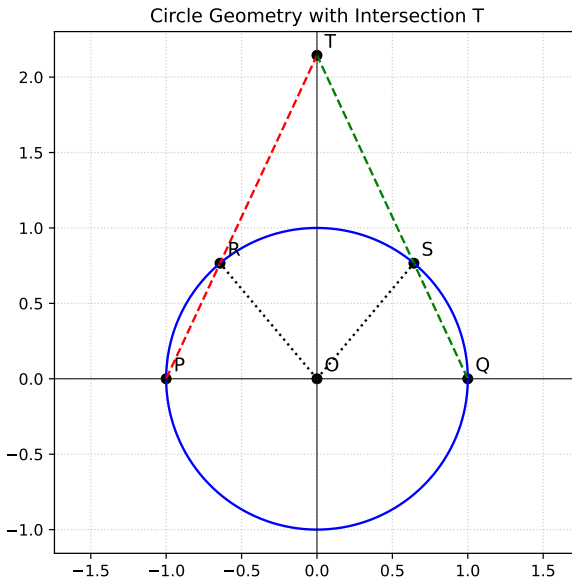


Fig. 1.1.2.1. Circle with diameter PQ , $\angle ROS = 80^\circ$ and $\angle RTS = 50^\circ$

codes/ce1/q07.py

figs/q07.pdf

1.1.4 P and Q are two positive integers such that $P^2 = Q^2 + 13$. The product of the numbers P and Q is _____

- [illegible]

Solution: The given equation can be written as

$$P^2 - Q^2 = (P - Q)(P + Q) = 13 \quad (1.1.4.1)$$

Since 13 is prime and $0 < P - Q < P + Q$,

$$P - Q = 1, \quad P + Q = 13 \quad (1.1.4.2)$$

Expressing in the matrix form (1.1.3.1):

$$\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} P \\ Q \end{pmatrix} = \begin{pmatrix} 1 \\ 13 \end{pmatrix} \quad (1.1.4.3)$$

Applying $R_2 \rightarrow R_2 - R_1$ on the augmented matrix:

$$\begin{pmatrix} 1 & -1 & 1 \\ 1 & 1 & 13 \end{pmatrix} \xrightarrow{R_2 \rightarrow R_2 - R_1} \begin{pmatrix} 1 & -1 & 1 \\ 0 & 2 & 12 \end{pmatrix} \implies Q = 6, P = 7 \quad (1.1.4.4)$$

Therefore,

$$PQ = 7 \times 6 = 42 \quad (1.1.4.5)$$

codes/ce1/q10.py

1.1.5 Matrix $\mathbf{P}$ is given as

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \quad (1.1.5.1)$$

The TRUE option is

- a) Trace of $\mathbf{P}$ is equal to the sum of the Eigen values of $\mathbf{P}$.
b) $\mathbf{P}^T \mathbf{P}$ is an identity matrix.
c) $\mathbf{P}$ is a skew-symmetric matrix.
d) Absolute magnitude of each Eigen value is 1.

Solution: From (1.1.4.1), expanding along the second row,

$$|\lambda \mathbf{I} - \mathbf{P}| = \begin{vmatrix} \lambda - 1 & 0 & -1 \\ 0 & \lambda - 1 & 0 \\ -1 & 0 & \lambda - 1 \end{vmatrix} \quad (1.1.5.2)$$

$$= (\lambda - 1) [(\lambda - 1)^2 - 1] \quad (1.1.5.3)$$

$$= \lambda (\lambda - 1) (\lambda - 2) = 0 \quad (1.1.5.4)$$

$$\implies \lambda_1 = 0, \lambda_2 = 1, \lambda_3 = 2 \quad (1.1.5.5)$$

From (1.1.4.2):

$$\text{tr}(\mathbf{P}) = 1 + 1 + 1 = 3 = \lambda_1 + \lambda_2 + \lambda_3 \quad (1.1.5.6)$$

so option (A) is true. The remaining options are false since

$$\mathbf{P}^\top \mathbf{P} = \begin{pmatrix} 2 & 0 & 2 \\ 0 & 1 & 0 \\ 2 & 0 & 2 \end{pmatrix} \neq \mathbf{I} \quad (1.1.5.7)$$

$$\mathbf{P}^\top = \mathbf{P} \neq -\mathbf{P} \quad (1.1.5.8)$$

$$|\lambda_1| = 0 \neq 1 \quad (1.1.5.9)$$

codes/ce1/q11.py

1.1.6 Given:

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (1.1.6.1)$$

The above system of equations represents a

- | | |
|----------|-----------|
| a) plane | c) volume |
| b) line | d) point |

Solution: The augmented matrix is reduced as

$$\begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 0 & 2 & 0 \end{pmatrix} \xrightarrow{R_2 \rightarrow R_2 - R_1} \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & -1 & 1 & 0 \end{pmatrix} \quad (1.1.6.2)$$

Thus,

$$\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A} \mid \mathbf{b}) = 2 < n = 3 \quad (1.1.6.3)$$

From (1.1.3.2), the system is consistent with $n - r = 1$ free variable. From (1.1.3.4), the solution is

$$\mathbf{x} = \kappa \left(\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \right) = \kappa \begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix}, \quad \kappa \in \mathbb{R} \quad (1.1.6.4)$$

which is a line through the origin (the intersection of two planes).

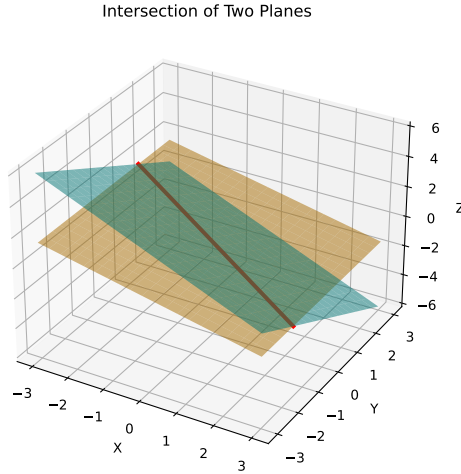


Fig. 1.1.6.1. The two planes and their line of intersection

codes/ce1/q12.py

figs/q12.pdf

1.1.7 Bag I contains 4 white and 6 black balls. Bag II contains 4 white and 3 black balls. One ball is drawn at random from any one of the two bags and it is found to be a black ball. The probability that the black ball was drawn from Bag I is _____ (rounded off to two decimal places).

Solution: Let B_1 , B_2 denote choosing Bag I, Bag II and E the event of drawing a black ball. The prior and likelihood vectors are

$$\pi = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}, \quad \mathbf{1} = \begin{pmatrix} \frac{6}{10} \\ \frac{3}{7} \end{pmatrix} \quad (1.1.7.1)$$

From (1.1.6.1),

$$\Pr(E) = \pi^\top \mathbf{1} = \frac{1}{2} \left(\frac{3}{5} + \frac{3}{7} \right) = \frac{18}{35} \quad (1.1.7.2)$$

From (1.1.6.2),

$$\Pr(B_1 | E) = \frac{\frac{1}{2} \times \frac{3}{5}}{\frac{18}{35}} = \frac{3}{10} \times \frac{35}{18} = \frac{7}{12} \approx 0.58 \quad (1.1.7.3)$$

codes/ce1/q26.py

1.1.8 A matrix is given as:

$$\begin{pmatrix} 9 & 15 \\ 15 & 50 \end{pmatrix} \quad (1.1.8.1)$$

By performing Cholesky decomposition, $|l_{22}|$ of the lower triangular matrix is _____ (in integer).

Solution: Comparing with (1.1.5.1),

$$l_{11}^2 = 9 \implies l_{11} = 3 \quad (1.1.8.2)$$

$$l_{11}l_{21} = 15 \implies l_{21} = 5 \quad (1.1.8.3)$$

$$l_{21}^2 + l_{22}^2 = 50 \implies l_{22}^2 = 25 \implies |l_{22}| = 5 \quad (1.1.8.4)$$

Thus,

$$\mathbf{L} = \begin{pmatrix} 3 & 0 \\ 5 & 5 \end{pmatrix}, \quad \mathbf{L}\mathbf{L}^\top = \begin{pmatrix} 9 & 15 \\ 15 & 50 \end{pmatrix} \quad (1.1.8.5)$$

codes/ce1/q27.py

1.1.9 It is given that x and y are integers in the following equation:

$$(x + y - 7)^2 + (y + 3x - 13)^2 = 0 \quad (1.1.9.1)$$

The value of $(x^3 + y^3)$ is _____ (in integer).

Solution: Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 3 & 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 7 \\ 13 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix} \quad (1.1.9.2)$$

The given equation is

$$\|\mathbf{Ax} - \mathbf{b}\|^2 = 0 \quad (1.1.9.3)$$

From (1.1.3.5),

$$\mathbf{Ax} = \mathbf{b} \quad (1.1.9.4)$$

Reducing the augmented matrix:

$$\begin{pmatrix} 1 & 1 & 7 \\ 3 & 1 & 13 \end{pmatrix} \xrightarrow{R_2 \rightarrow R_2 - 3R_1} \begin{pmatrix} 1 & 1 & 7 \\ 0 & -2 & -8 \end{pmatrix} \implies y = 4, x = 3 \quad (1.1.9.5)$$

Therefore,

$$x^3 + y^3 = 27 + 64 = 91 \quad (1.1.9.6)$$

codes/ce1/q28.py

figs/q28.pdf

1.1.10 Let $f(x)$ be a continuous function defined in $[0, 2] \rightarrow \mathbb{R}$ and satisfying the equation

$$\int_0^2 f(x) [x - f(x)] dx = \frac{2}{3} \quad (1.1.10.1)$$

The value of $f(1)$ is

- a) 1
b) 2
c) $\frac{1}{2}$
d) 0

Solution: Using (1.1.7.1) with $a = x$ and $b = f(x)$:

$$f(x) [x - f(x)] = \frac{x^2}{4} - \left(f(x) - \frac{x}{2}\right)^2 \quad (1.1.10.2)$$

Integrating over $[0, 2]$:

$$\int_0^2 f(x) [x - f(x)] dx = \int_0^2 \frac{x^2}{4} dx - \int_0^2 \left(f(x) - \frac{x}{2}\right)^2 dx \quad (1.1.10.3)$$

$$= \frac{2}{3} - \int_0^2 \left(f(x) - \frac{x}{2}\right)^2 dx \quad (1.1.10.4)$$

Equating with the given value,

$$\int_0^2 \left(f(x) - \frac{x}{2}\right)^2 dx = 0 \quad (1.1.10.5)$$

Since the integrand is continuous and non-negative,

$$f(x) = \frac{x}{2} \implies f(1) = \frac{1}{2} \quad (1.1.10.6)$$

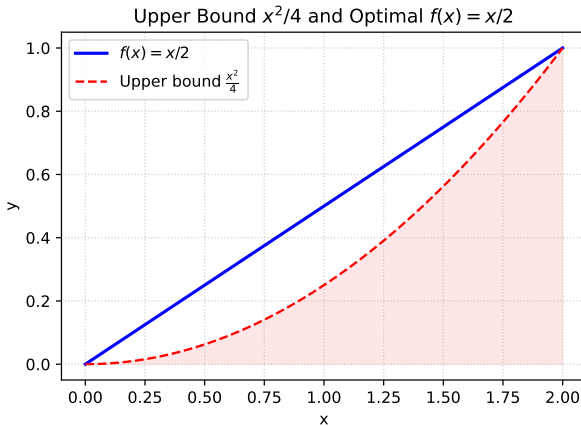


Fig. 1.1.10.1. The integrand attains its upper bound $\frac{x^2}{4}$ only for $f(x) = \frac{x}{2}$

codes/ce1/q36.py

figs/q36.pdf

1.1.11 An ordinary differential equation is given below.

$$x^2 \frac{d^2 y}{dx^2} = 6y \quad (1.1.11.1)$$

Considering a and b as arbitrary constants, the general solution of the equation is

$$\begin{array}{ll} \text{a) } y(x) = ax^3 + \frac{b}{x^2} & \text{c) } y(x) = ax^2 + b \ln x \\ \text{b) } y(x) = ax^2 + \frac{b}{x^3} & \text{d) } y(x) = ax^3 + b \ln x \end{array}$$

Solution: Let $x = e^t$ ($x > 0$). Then

$$x \frac{dy}{dx} = \frac{dy}{dt}, \quad x^2 \frac{d^2 y}{dx^2} = \frac{d^2 y}{dt^2} - \frac{dy}{dt} \quad (1.1.11.2)$$

and the equation becomes

$$y'' - y' - 6y = 0 \quad (1.1.11.3)$$

Taking the Laplace transform using (1.1.8.1) and (1.1.8.2):

$$s^2 Y(s) - sy(0) - y'(0) - [sY(s) - y(0)] - 6Y(s) = 0 \quad (1.1.11.4)$$

$$Y(s) = \frac{(s-1)y(0) + y'(0)}{s^2 - s - 6} = \frac{(s-1)y(0) + y'(0)}{(s-3)(s+2)} \quad (1.1.11.5)$$

Decomposing into partial fractions and using (1.1.8.3),

$$Y(s) = \frac{a}{s-3} + \frac{b}{s+2} \implies y(t) = ae^{3t} + be^{-2t} \quad (1.1.11.6)$$

Equivalently, from (1.1.8.4), the exponents are the eigenvalues of

$$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 6 & 1 \end{pmatrix}, \quad |\lambda \mathbf{I} - \mathbf{A}| = \lambda^2 - \lambda - 6 = 0 \implies \lambda = 3, -2 \quad (1.1.11.7)$$

Substituting $e^t = x$:

$$y(x) = ax^3 + \frac{b}{x^2} \quad (1.1.11.8)$$

codes/ce1/q37.py

1.1.12 A plane truss consists of two linearly elastic, homogeneous, identical members, namely PQ and QR . Both members have length (L), cross-sectional area (A), and modulus of elasticity (E). The members are inclined at 45° as shown in the figure. The truss has hinge supports at P and R . The translational degrees-of-freedom (u and v) are shown at joint Q .

After application of the boundary conditions, the stiffness matrix of the truss becomes:

$$\text{a) } \frac{AE}{L} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

$$\text{b) } \frac{AE}{L} \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$$

$$\text{c) } \frac{AE}{L} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\text{d) } \frac{AE}{L} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

Solution: The unit direction vectors of the members are

$$\mathbf{m}_{PQ} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{m}_{QR} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (1.1.12.1)$$

From (1.1.12.1),

$$\mathbf{k}_{PQ} = \frac{AE}{L} \mathbf{m}_{PQ} \mathbf{m}_{PQ}^T = \frac{AE}{L} \begin{pmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{pmatrix} \quad (1.1.12.2)$$

$$\mathbf{k}_{QR} = \frac{AE}{L} \mathbf{m}_{QR} \mathbf{m}_{QR}^T = \frac{AE}{L} \begin{pmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{pmatrix} \quad (1.1.12.3)$$

After deleting the restrained degrees of freedom at P and R , only the (u, v) block at Q receives contributions from both members:

$$\mathbf{K} = \mathbf{k}_{PQ} + \mathbf{k}_{QR} = \frac{AE}{L} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (1.1.12.4)$$

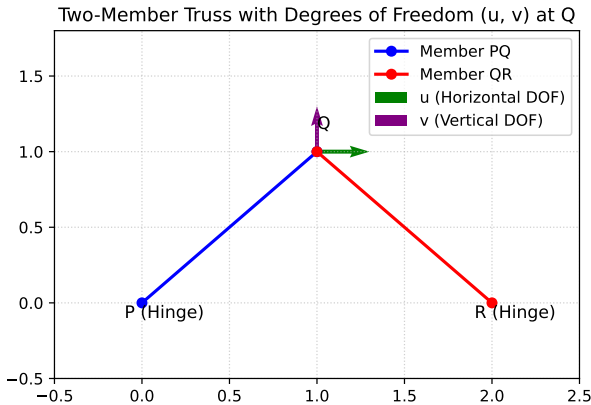


Fig. 1.1.12.1. Two-member truss with degrees of freedom u, v at Q

codes/ce1/q38.py

figs/q38.pdf

1.1.13 The plane frame has a hinge and a roller support, and is loaded as shown in the figure. Both the columns have same height.

What is the absolute value of the maximum bending moment (in kN-m) in the frame?

a) 165

c) 240

b) 150

d) 195

Solution: From the figure, the frame has a hinge at A , a roller at B , columns of height 3 m and a beam CD of span 4 m. A horizontal load of 50 kN acts at C and a vertical load of 90 kN acts at the mid-point E of CD :

$$\mathbf{A} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \mathbf{C} = \begin{pmatrix} 0 \\ 3 \end{pmatrix}, \mathbf{E} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}, \mathbf{D} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 4 \\ 0 \end{pmatrix}, \mathbf{F}_C = \begin{pmatrix} 50 \\ 0 \end{pmatrix}, \mathbf{F}_E = \begin{pmatrix} 0 \\ -90 \end{pmatrix} \quad (1.1.13.1)$$

The reactions are $\mathbf{R}_A = (H_A \ V_A)^\top$ and $\mathbf{R}_B = (0 \ V_B)^\top$. From (1.1.11.1) with moments about $\mathbf{A}$,

$$(\mathbf{C} - \mathbf{A}) \times \mathbf{F}_C = 0 \times 0 - 3 \times 50 = -150 \quad (1.1.13.2)$$

$$(\mathbf{E} - \mathbf{A}) \times \mathbf{F}_E = 2 \times (-90) - 3 \times 0 = -180 \quad (1.1.13.3)$$

$$(\mathbf{B} - \mathbf{A}) \times \mathbf{R}_B = 4V_B \quad (1.1.13.4)$$

In matrix form:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 4 \end{pmatrix} \begin{pmatrix} H_A \\ V_A \\ V_B \end{pmatrix} = \begin{pmatrix} -50 \\ 90 \\ 330 \end{pmatrix} \Rightarrow \begin{pmatrix} H_A \\ V_A \\ V_B \end{pmatrix} = \begin{pmatrix} -50 \\ 7.5 \\ 82.5 \end{pmatrix} \text{ kN} \quad (1.1.13.5)$$

Using (1.1.11.2), the bending moment in column AC at height y (forces below the section) is

$$|M| = \left| \left(\mathbf{A} - \begin{pmatrix} 0 \\ y \end{pmatrix} \right) \times \mathbf{R}_A \right| = 50y \Rightarrow |M_C| = 150 \text{ kN m} \quad (1.1.13.6)$$

and in the beam at E (forces to the right of the section) is

$$|M_E| = |(\mathbf{B} - \mathbf{E}) \times \mathbf{R}_B| = |2 \times 82.5 - (-3) \times 0| = 165 \text{ kN m} \quad (1.1.13.7)$$

In column BD , the roller reaction is vertical and along the column, so $M = 0$. The maximum bending moment is 165 kN m under the 90 kN load.

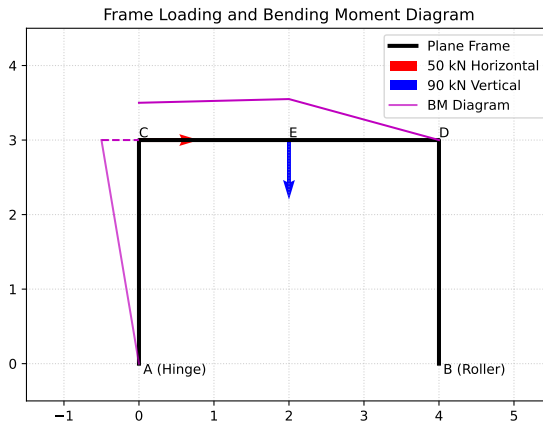


Fig. 1.1.13.1. Frame loading and bending moment diagram

codes/ce1/q39.py

figs/q39.pdf

1.1.14 Starting with the first approximation as $x = 0.5$, the second approximation for the root of the following function by the Newton-Raphson method is _____ (rounded off to two decimal places).

$$f(x) = e^{-x} - x \quad (1.1.14.1)$$

Solution: The function and its derivative are

$$f(x) = e^{-x} - x \quad (1.1.14.2)$$

$$f'(x) = -e^{-x} - 1 \quad (1.1.14.3)$$

Substituting into (1.1.9.1):

$$x_{n+1} = x_n + \frac{e^{-x_n} - x_n}{e^{-x_n} + 1} \quad (1.1.14.4)$$

$$= \frac{1 + x_n}{1 + e^{x_n}} \quad (1.1.14.5)$$

With the first approximation $x_0 = 0.5$, the second approximation is

$$x_1 = \frac{1.5}{1 + e^{0.5}} = \frac{1.5}{2.64872} = 0.56631 \approx 0.57 \quad (1.1.14.6)$$

Subsequent iterations converge to $x_2 = x_3 = 0.567143$.

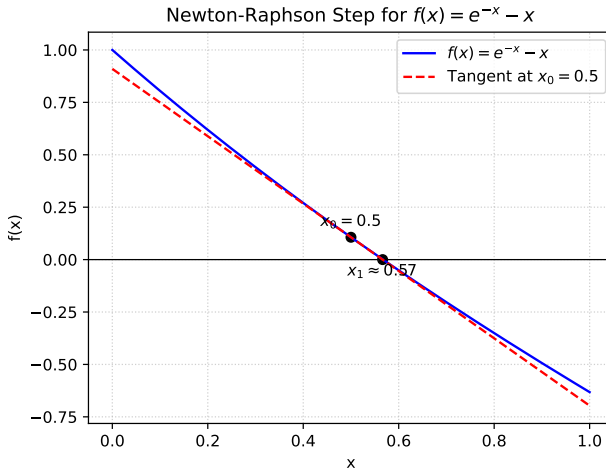


Fig. 1.1.14.1. One Newton-Raphson step for $f(x) = e^{-x} - x$ from $x_0 = 0.5$

codes/ce1/q48.py

figs/q48.pdf

1.1.15 Values of y for different values of x are tabulated below.

x	-2	1	2
y	28	4	16

If a second-degree interpolating polynomial $P_2(x)$ is used to represent y , the value of $P_2(0)$ is _____ (rounded off to the nearest integer).

Solution: Let $\mathbf{c} = (c_0 \ c_1 \ c_2)^\top$ and $\mathbf{y} = (28 \ 4 \ 16)^\top$. From (1.1.10.1),

$$\begin{pmatrix} 1 & -2 & 4 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{pmatrix} \mathbf{c} = \mathbf{y} \quad (1.1.15.1)$$

Reducing the augmented matrix:

$$\begin{pmatrix} 1 & -2 & 4 & 28 \\ 1 & 1 & 1 & 4 \\ 1 & 2 & 4 & 16 \end{pmatrix} \xrightarrow[R_3 \rightarrow R_3 - R_1]{R_2 \rightarrow R_2 - R_1} \begin{pmatrix} 1 & -2 & 4 & 28 \\ 0 & 3 & -3 & -24 \\ 0 & 4 & 0 & -12 \end{pmatrix} \quad (1.1.15.2)$$

$$\Rightarrow c_1 = -3, \ c_2 = 5, \ c_0 = 2 \quad (1.1.15.3)$$

Thus,

$$\mathbf{c} = \begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix} \Rightarrow P_2(x) = 5x^2 - 3x + 2 \quad (1.1.15.4)$$

and

$$P_2(0) = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} \mathbf{c} = 2 \quad (1.1.15.5)$$

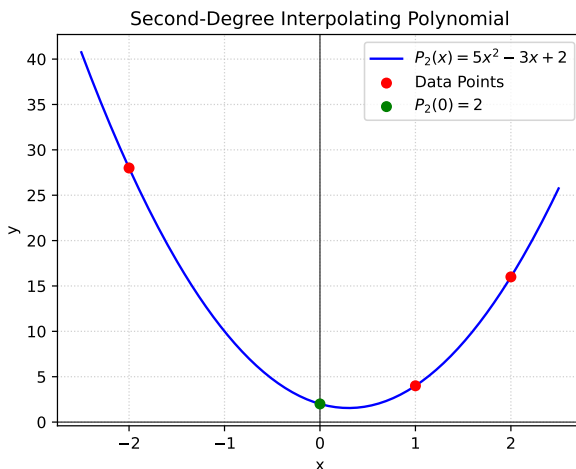


Fig. 1.1.15.1. Interpolating polynomial through the given points

codes/ce1/q49.py

figs/q49.pdf

- 1.1.16 Two identical blocks A and B are connected by a rigid rod. The blocks rest against vertical and horizontal planes, as shown in the figure. The coefficient of static friction at the vertical and horizontal planes is the same. If the sliding impends when $\theta = 45^\circ$, the value of the coefficient of static friction is _____ (rounded off to two decimal places).

Solution: Let each block have weight W and let the rod make angle θ with the horizontal, with $c = \cos \theta$, $s = \sin \theta$. The rod is a two-force member carrying compression C along

$$\mathbf{e} = \frac{\mathbf{A} - \mathbf{B}}{\|\mathbf{A} - \mathbf{B}\|} = \begin{pmatrix} -c \\ s \end{pmatrix} \quad (1.1.16.1)$$

When sliding impends, A moves down and B moves away from the wall, so the friction forces are μN_A (upward on A) and μN_B (towards the wall on B). From

(1.1.11.1):

$$\text{Block } A : \begin{pmatrix} N_A \\ \mu N_A \end{pmatrix} + \begin{pmatrix} 0 \\ -W \end{pmatrix} + C \begin{pmatrix} -c \\ s \end{pmatrix} = \mathbf{0} \quad (1.1.16.2)$$

$$\text{Block } B : \begin{pmatrix} -\mu N_B \\ N_B \end{pmatrix} + \begin{pmatrix} 0 \\ -W \end{pmatrix} - C \begin{pmatrix} -c \\ s \end{pmatrix} = \mathbf{0} \quad (1.1.16.3)$$

This gives four equations in three unknowns $(N_A \ N_B \ C)^\top$:

$$\begin{pmatrix} 1 & 0 & -c \\ \mu & 0 & s \\ 0 & -\mu & c \\ 0 & 1 & -s \end{pmatrix} \begin{pmatrix} N_A \\ N_B \\ C \end{pmatrix} = \begin{pmatrix} 0 \\ W \\ 0 \\ W \end{pmatrix} \quad (1.1.16.4)$$

From (1.1.3.3), the system is consistent only if

$$\begin{vmatrix} 1 & 0 & -c & 0 \\ \mu & 0 & s & W \\ 0 & -\mu & c & 0 \\ 0 & 1 & -s & W \end{vmatrix} = 0 \quad (1.1.16.5)$$

Expanding along the second column,

$$\mu \begin{vmatrix} 1 & -c & 0 \\ \mu & s & W \\ 0 & -s & W \end{vmatrix} + \begin{vmatrix} 1 & -c & 0 \\ \mu & s & W \\ 0 & c & 0 \end{vmatrix} = 0 \quad (1.1.16.6)$$

$$\mu W (2s + \mu c) - Wc = 0 \quad (1.1.16.7)$$

$$\implies \tan \theta = \frac{1 - \mu^2}{2\mu} \quad (1.1.16.8)$$

For $\theta = 45^\circ$:

$$\mu^2 + 2\mu - 1 = 0 \implies \mu = -1 + \sqrt{2} = 0.414 \quad (1.1.16.9)$$

taking the positive root.

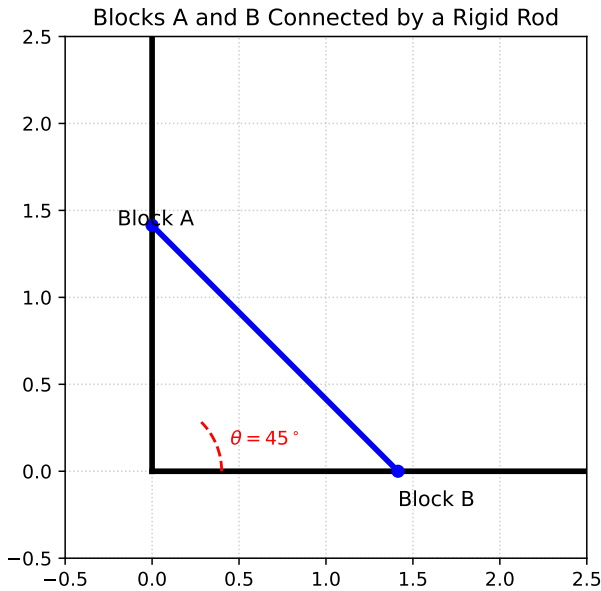


Fig. 1.1.16.1. Blocks *A* and *B* connected by a rigid rod at $\theta = 45^\circ$

codes/ce1/q50.py

figs/q50.pdf

1.1.17 A homogenous, linearly elastic rod *AB* is connected to a linearly elastic spring *BC* in between the fixed supports at *A* and *C*, as shown in the figure. The cross-sectional area, modulus of elasticity, and the coefficient of thermal expansion of the rod *AB* are 500 mm^2 , $60 \times 10^3 \text{ MPa}$, and $12 \times 10^{-6} \text{ per } ^\circ\text{C}$, respectively. The stiffness (*k*) of spring *BC* is 2500 N/mm .

The internal force (in kN) that will develop in the spring *BC* when the temperature of rod *AB* is increased by 100°C is _____ (rounded off to one decimal place).

Solution: From the figure, the length of rod *AB* is $L = 3 \text{ m} = 3000 \text{ mm}$. The rod stiffness is

$$k_r = \frac{AE}{L} = \frac{500 \times 60 \times 10^3}{3000} = 10^4 \text{ N/mm} = 10000 \text{ N/mm} \quad (1.1.17.1)$$

The free thermal expansion of the rod is

$$\delta_{\text{th}} = L\alpha\Delta T = 3000 \times 12 \times 10^{-6} \times 100 = 3.6 \text{ mm} \quad (1.1.17.2)$$

Since the rod and spring are in series between fixed supports, the internal compressive force F satisfies

$$\frac{F}{k_r} + \frac{F}{k_s} = \delta_{th} \implies F = \frac{\delta_{th}}{\frac{1}{k_r} + \frac{1}{k_s}} \quad (1.1.17.3)$$

Substituting $k_r = 10000 \text{ N/mm}$ and $k_s = 2500 \text{ N/mm}$:

$$F = \frac{3.6}{\frac{1}{10000} + \frac{1}{2500}} = \frac{3.6}{0.0005} = 7200 \text{ N} = 7.2 \text{ kN} \quad (1.1.17.4)$$

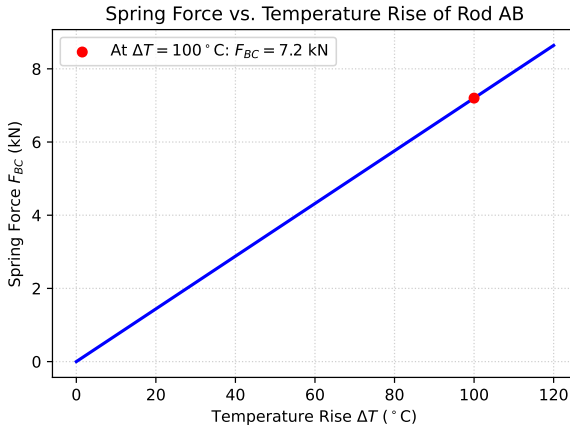


Fig. 1.1.17.1. Rod AB connected in series with spring BC under thermal loading

codes/ce1/q51.py

figs/q51.pdf